



**BITS Pilani**  
Pilani | Dubai | Goa | Hyderabad

# Introduction to Statistical Methods

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**ISM Team**



## Baye's theorem and Naïve Bayes theorem

# Revise

- Descriptive Statistics
- Review of Set Theory, Counting Principles
- Probability, Axioms of Probability
- Addition and Multiplication Rule of Probability
- Conditional Probability and Independent events
- Total Probability

# Contact Session 4

| Contact Session | List of Topic Title                                            | Reference |
|-----------------|----------------------------------------------------------------|-----------|
| CS - 4          | Bayes theorem(with proof),Introduction to Naïve Bayes concept. | T1 & T2   |
| HW              | Problems on Bayes theorem                                      | T1 & T2   |



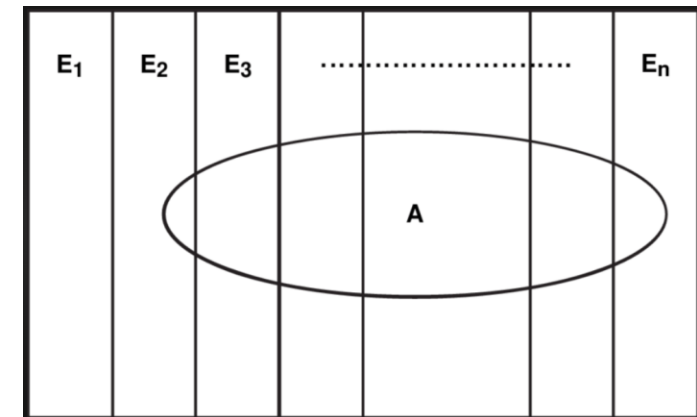
# BAYE'S THEOREM:

Suppose that  $E_1, E_2, \dots, E_n$  are mutually exclusive events of a sample space “S”

such that  $P(E_i) > 0$  for  $i = 1, 2, 3, \dots, n$  and  $A$  is any arbitrary event of “S”

such that  $P(A) > 0$  and  $A \subseteq \bigcup_{i=1}^n E_i$  then the conditional probability of  $E_i$  given  $A$

for  $i = 1, 2, 3, \dots, n$  is given by 
$$P(E_i/A) = \frac{P(E_i)P(A/E_i)}{\sum_{i=1}^n P(E_i)P(A/E_i)}$$



# Proof:

Given that  $A \subseteq \bigcup_{i=1}^n E_i$

$$\therefore A = A \cap \left( \bigcup_{i=1}^n E_i \right)$$

$$\Rightarrow A = A \cap (E_1 \cup E_2 \cup \dots \cup E_n)$$

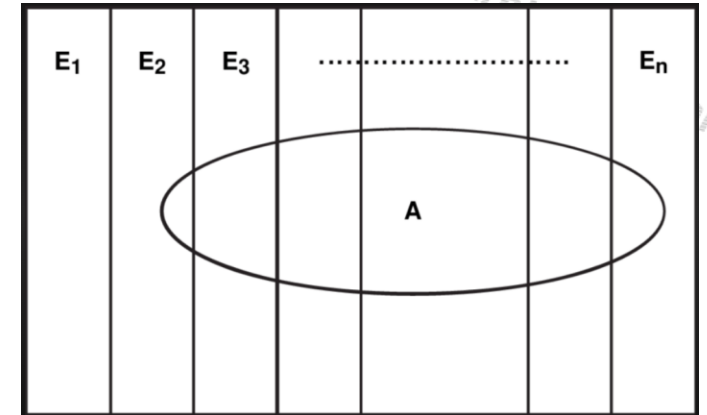
$$A = (A \cap E_1) \cup (A \cap E_2) \dots \cup (A \cap E_n) \quad (\text{Distributive law})$$

$$\text{ie } A = \bigcup_{i=1}^n (A \cap E_i)$$

Since  $E_1, E_2, \dots, E_n$  are mutually exclusive events,  $E_i \cap E_j = \emptyset$  for  $i \neq j$

and  $A \cap E_1, A \cap E_2, \dots, A \cap E_n$  are also mutually exclusive

$$(A \cap E_i) \cap (A \cap E_j) = \emptyset \quad \text{for } i \neq j$$



## Proof:(Continuation)

Applying the addition rule for mutually exclusive events, we have

$$\begin{aligned} P(A) &= P((A \cap E_1) \cup (A \cap E_2) \dots \cup (A \cap E_n)) \\ &= P(A \cap E_1) + P(A \cap E_2) + \dots + P(A \cap E_n) \end{aligned}$$

$$P(A) = P\left(\bigcup_{i=1}^n (A \cap E_i)\right) = \sum_{i=1}^n P(A \cap E_i) \quad \text{--- (1)}$$

From multiplication rule we have

$$P(A \cap E_i) = P(E_i) P(A/E_i) \quad \text{----- (2)} \quad \text{Substituting (2) in (1) we have}$$

$$P(A) = \sum_{i=1}^n P(E_i) P(A/E_i) \quad \text{--- (3) [Rule of total probability OR Rule of elimination]}$$

## Proof:(Continuation)

- From the definition of conditional probability

$$P(E_i / A) = \frac{P(E_i \cap A)}{P(A)} \text{ ----- (4)}$$

- Substituting (2) and (3) in (4) we get

$$P(E_i / A) = \frac{P(E_i)P(A/E_i)}{\sum_i^n P(E_i)P(A/E_i)}$$

## Example:1

The Probabilities of  $X, Y$  and  $Z$  becoming managers are  $\frac{4}{9}, \frac{2}{9}$ , and  $\frac{1}{3}$  respectively.

The probabilities that the bonus schemes will be introduced if  $X, Y$  and  $Z$  becomes managers are  $\frac{3}{10}, \frac{1}{2}$  and  $\frac{4}{5}$  respectively.

- (i) What is the probability that bonus scheme will be introduced ?
- (ii) If the bonus scheme has been introduced, What is the probability that the manager appointed was  $X$ ?

## Solution:

Let  $E_1, E_2, E_3$  denote the events that X, Y and Z become Managers respectively and A denote the event that Bonus scheme is introduced.

$$P(E_1) = \frac{4}{9}, P(E_2) = \frac{2}{9}, P(E_3) = \frac{1}{3}, P(A|E_1) = \frac{3}{10}, P(A|E_2) = \frac{1}{2}, P(A|E_3) = \frac{4}{5}$$

(i) Using Addition theorem of Probability

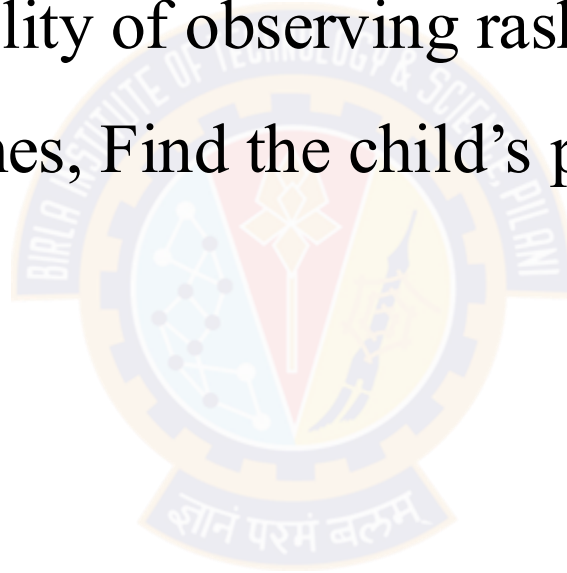
$$\begin{aligned} P(A) &= P(E_1 \cap A) + P(E_2 \cap A) + P(E_3 \cap A) \\ &= P(E_1)P(A|E_1) + P(E_2)P(A|E_2) + P(E_3)P(A|E_3) = \frac{4}{9} \times \frac{3}{10} + \frac{2}{9} \times \frac{1}{2} + \frac{1}{3} \times \frac{4}{5} = \frac{23}{45} \end{aligned}$$

(ii) Using Baye's Theorem,

$$P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(A)} = \frac{6}{23}$$

## Example:2

In a neighborhood, 90% children were falling sick due to flu and 10% due to measles and no other disease. The probability of observing rashes for measles is 0.95 and for flu is 0.08. If a child develops rashes, Find the child's probability of having flu.



## Solution:

Let, F: children with flu M: children with measles

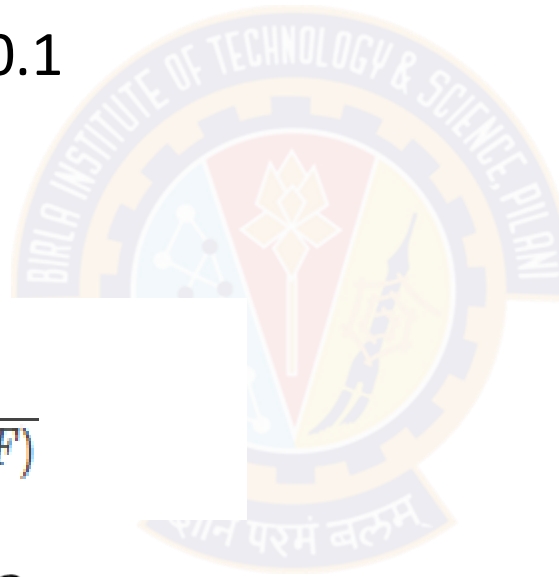
R: children showing the symptom of rash

$$P(F) = 90\% = 0.9, P(M) = 10\% = 0.1$$

$$P(R|F) = 0.08, P(R|M) = 0.95$$

$$P(F|R) = \frac{P(R|F)P(F)}{P(R|M)P(M) + P(R|F)P(F)}$$

$$P(F|R) = \frac{0.08 \times 0.9}{0.95 \times 0.1 + 0.08 \times 0.9} = 0.43$$



## Example:3

It is estimated that 50% of emails are spam emails. Some software has been applied to filter these spam emails before they reach your inbox. A certain brand of software claims that it can detect 99% of spam e-mails, and the probability for a false positive (a non-spam email detected as spam) is 5%. Now if an email is detected as spam, then what is the probability that it is in fact a non-spam email?

## Solution:

Let  $A$  = event that an email is detected as spam,  $B$  = event that an email is spam,  
 $B^c$  = event that an email is not spam.

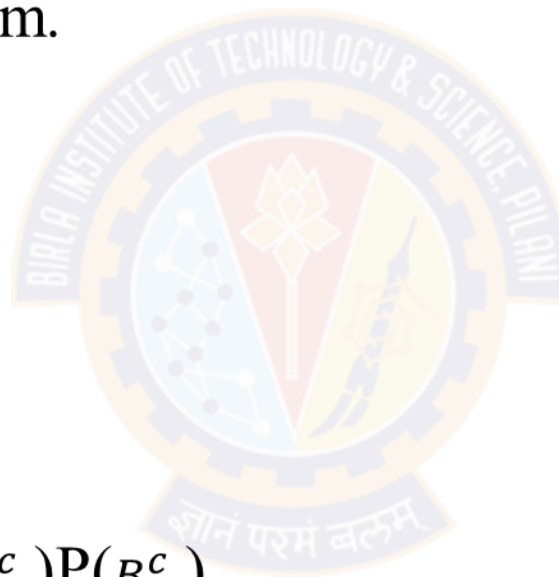
We know  $P(B) = P(B^c) = .5$ ,

$P(A | B) = 0.99$ ,  $P(A | B^c) = 0.05$ .

Hence by the Bayes's formula

$$\text{we have } P(B^c | A) = \frac{P(A | B^c)P(B^c)}{P(A | B)P(B) + P(A | B^c)P(B^c)}$$

$$= (0.05 \times 0.5) / (0.05 \times 0.5 + 0.99 \times 0.5) = \frac{5}{104}$$



# Bayes' Theorem for Hypothesis

$$P(H | E) = \frac{P(E | H)P(H)}{P(E)}$$

Where

$P(H | E)$  : Posterior probability of hypothesis  $H$  given evidence  $E$ .

$P(H)$  : Prior probability of hypothesis  $H$ .

$P(E | H)$  : Likelihood of evidence  $E$  given  $H$ .

$P(E)$  : Evidence probability, which acts as a normalization constant.

If  $H_1, H_2, \dots, H_n$  are **mutually exclusive and exhaustive hypotheses**, the sum of their posterior probabilities is always 1 :  $\sum_{i=1}^n P(H_i | E) = 1$

This holds because of the total probability of  $E$  accounts for all hypotheses.

This logic forms the core of the **Naive Bayes classifier**, widely used in:

- Spam filtering
- Sentiment analysis
- Document categorization
- Medical diagnosis prediction

## Example 4

### Email Spam Classification using Bayes' Theorem

- A machine learning model needs to decide whether an email is **Spam ( $H_1$ )** or **Not Spam ( $H_2$ )** based on the presence of the word “offer” (E).
- **Given Data (from training corpus):**
  - $P(\text{Spam}) = 0.3 \rightarrow 30\%$  of all emails are spam
  - $P(\text{Not Spam}) = 0.7 \rightarrow 70\%$  of all emails are not spam
  - $P(\text{“offer”} \mid \text{Spam}) = 0.8 \rightarrow 80\%$  of spam emails contain the word “offer”
  - $P(\text{“offer”} \mid \text{Not Spam}) = 0.1 \rightarrow 10\%$  of non-spam emails contain “offer”
  - **Find the probability that an email is spam given that it contains “offer”.**

## Example 4: Solution

Using Bayes' Theorem:

$$P(\text{Spam} / \text{Offer}) = \frac{P(\text{Offer} / \text{Spam})P(\text{Spam})}{P(\text{Offer})}$$

Where  $P(\text{Offer}) = P(\text{Offer} / \text{Spam})P(\text{Spam}) + P(\text{Offer} / \text{Not Spam})P(\text{Not Spam})$

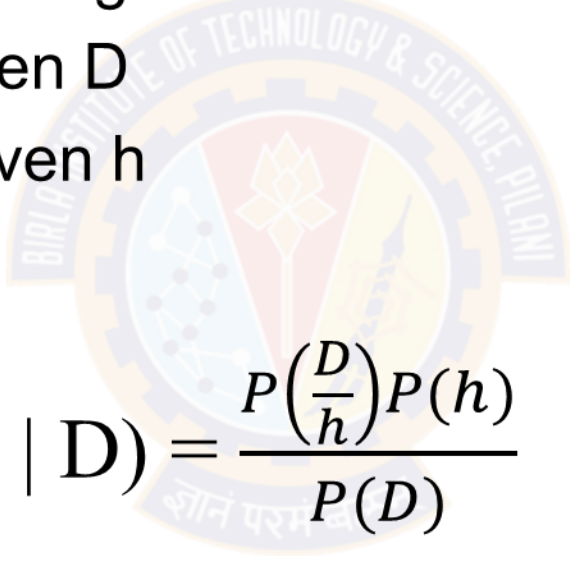
$$= (0.8)(0.3) + (0.1)(0.7) = 0.24 + 0.07 = 0.31$$

$$P(\text{Spam} / \text{Offer}) = \frac{0.8 \times 0.3}{0.31} = 0.774$$

If an email contains the word “offer”, there is a **77.4% probability** that it is **spam**.

# Baye's Theorem

- $P(h)$  = prior probability of hypothesis  $h$ , before seeing the training data
- $P(D)$  = prior probability of training data  $D$
- $P(h \mid D)$  = probability of  $h$  given  $D$
- $P(D \mid h)$  = probability of  $D$  given  $h$


$$P(h \mid D) = \frac{P\left(\frac{D}{h}\right)P(h)}{P(D)}$$

# Features of Bayesian learning

- Prior knowledge can be combined with observed data to determine the final probability of a hypothesis.
- Prior knowledge is provided by asserting
  - ❖ prior probability for each candidate hypothesis, and
  - ❖ probability distribution over observed data for each possible hypothesis.
- New instances can be classified by combining the predictions of multiple hypothesis, weighted by their probabilities.

# Choosing hypothesis

Generally want the most probable hypothesis given the training data  
Maximum a posteriori hypothesis  $h_{MAP}$ :

$$h_{MAP} = \arg \max_{h \in H} P(h|D)$$

$$= \arg \max_{h \in H} \frac{P(D|h).P(h)}{P(D)}$$

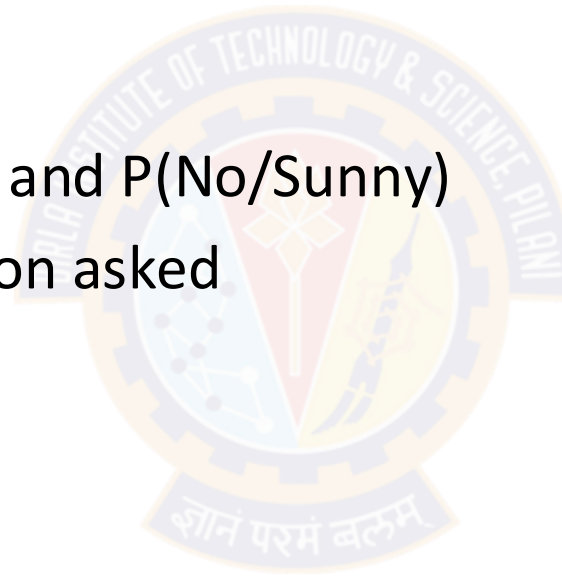
$$= \arg \max_{h \in H} P(D|h).P(h)$$

If assume  $P(h_i) = P(h_j)$  then can further simplify, and choose the Maximum likelihood (ML) hypothesis

$$h_{ML} = \arg \max_{h_i \in H} P(D|h_i)$$

## Bayes Theorem: Example : 5

- If the weather is sunny, Then the player will play or not?  
i.e.,  $\text{Play} / \text{sunny} = \text{Yes or No.}$
- Note: If we know  $P(\text{Yes/Sunny})$  and  $P(\text{No/Sunny})$
- then we can answer the question asked



| Weather  | Play |
|----------|------|
| Sunny    | No   |
| Overcast | Yes  |
| Rainy    | Yes  |
| Sunny    | Yes  |
| Sunny    | Yes  |
| Overcast | Yes  |
| Rainy    | No   |
| Rainy    | No   |
| Sunny    | Yes  |
| Rainy    | Yes  |
| Sunny    | No   |
| Overcast | Yes  |
| Overcast | Yes  |
| Rainy    | No   |

# Steps to Apply Bayes Theorem

- Step1: View or Collect “raw” data
- Step2: Convert long data into frequency table

| Weather  | No | Yes |
|----------|----|-----|
| Sunny    | 2  | 3   |
| Overcast | 0  | 4   |
| Rainy    | 3  | 2   |

- Step3: Row and Column sums to get Probabilities
- Weather Probabilities: Sunny =5/14,Rainy=5/14,Overcast= 4/14
- Play Probabilities :
  - No =5/14
  - Yes =9/14

| Weather  | Play |
|----------|------|
| Sunny    | No   |
| Overcast | Yes  |
| Rainy    | Yes  |
| Sunny    | Yes  |
| Sunny    | Yes  |
| Overcast | Yes  |
| Rainy    | No   |
| Rainy    | No   |
| Sunny    | Yes  |
| Rainy    | Yes  |
| Sunny    | No   |
| Overcast | Yes  |
| Overcast | Yes  |
| Rainy    | No   |

continuation

| Weather      | No | Yes         | Row Total |                  |
|--------------|----|-------------|-----------|------------------|
| Sunny        | 2  | 3           | 5         | P(sunny)= 5/14   |
| Overcast     | 0  | 4           | 4         | P(overcast)=4/14 |
| Rainy        | 3  | 2           | 5         | P(Rainy)= 5/14   |
| Column Total | 5  | 9           | 14        |                  |
| P(no)=5/14   |    | P(yes)=9/14 |           |                  |

Step4: Apply Probabilities from frequency table to Bayes Theorem

$$P(\text{Yes} \mid \text{Sunny}) = \frac{P(\text{Sunny} \mid \text{yes})P(\text{yes})}{P(\text{Sunny})} = \frac{\frac{3}{9} \times \frac{9}{14}}{\frac{5}{14}} = 0.60, \quad P(\text{No} \mid \text{Sunny}) = \frac{P(\text{Sunny} \mid \text{No})P(\text{No})}{P(\text{Sunny})} = \frac{\frac{2}{5} \times \frac{5}{14}}{\frac{5}{14}} = 0.40$$

$P(\text{Yes} \mid \text{Sunny}) > P(\text{No} \mid \text{Sunny})$  . Hence, The Player will play the game.

# Bayesian Learning

- Naive Bayes is a set of simple and efficient machine learning algorithms for solving a variety of classification and regression problems.
- **Naive Bayes assumes conditional independence where Bayes theorem does not. This means the relationship between all input features are independent.**
- Bayesian learning algorithms that calculate explicit probabilities for hypotheses, such as the Naive Bayes classifier, are among the most practical approaches to certain types of learning problems.

Example:

Problem of learning to classify text documents such as electronic news articles.

- For such learning tasks, the Naive Bayes classifier is among the most effective known algorithms.

# Naive Bayes Classifier

- In Naive Bayes, we classify data based on conditional probabilities:

$$P(C | X) = \frac{P(X | C)P(C)}{P(X)}$$

- C: Class label (hypothesis).
- X: Feature vector (evidence).

- The sum of probabilities over all possible classes  $C_1, C_2, \dots, C_n$  is **always 1**:

$$\sum_{i=1}^n P(C_i | X) = 1$$

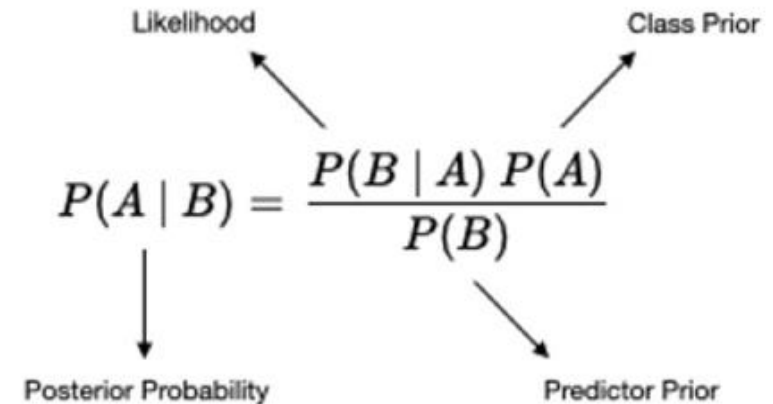
- This is because the Naive Bayes classifier calculates probabilities for a set of mutually exclusive and exhaustive classes. The denominator  $P(X)$  normalizes the probabilities, ensuring they sum to 1.

# Generative Model

- ❖ Build model to estimate the posterior probability  $P(Y | X)$  by estimating
- ❖ likelihood of data given target (hypothesis)  $P(X | Y)$
- ❖ Prior probabilities over target  $P(Y)$
- ❖ In general, for a specific class  $Y=c_k$ ,

$$P(Y = c_k | X) = \frac{P(X | Y = c_k) * P(Y = c_k)}{P(X)}$$

$$P(C | X) = \frac{P(X | C)P(C)}{P(X)}$$



# Conditional Independence

- Naive Bayes assumes conditional independence where Bayes theorem does not. This means the relationship between all input features are independent.
- **Definition:**  $X$  is conditionally independent of  $Y$  given  $Z$ , if the probability distribution governing  $X$  is independent of the value of  $Y$ , given the value of  $Z$ 
  - $(\forall i, j, k) P(X = x_i | Y = y_j, Z = z_k) = P(X = x_i | Z_k)$
  - $P(X|Y, Z) = P(X|Z)$
  - **Example: 1.**  $P(\text{Thunder} | \text{Rain}, \text{Lightning}) = P(\text{Thunder} | \text{Lightning})$
  - **2.**  $P(\text{flu} | \text{fever}, \text{blood test for flu}) = P(\text{flu} | \text{blood test for flu})$

## Applying Conditional Independence

**Naïve Bayes assumes  $X_i$  are conditionally independent given  $Y$**

e.g.,  $P(X_1|X_2, Y) = P(X_1|Y)$

$$\begin{aligned} P(X_1, X_2|Y) &= P(X_1|X_2, Y)P(X_2|Y) \\ &= P(X_1|Y)P(X_2|Y) \end{aligned}$$

General form:  $P(X_1, \dots, X_n|Y) = \prod_{j=1}^n P(X_j|Y)$

How many parameters to describe  $P(X_1, \dots, X_n | Y)$ ?  $P(Y)$ ?

Without conditional independence assumption?

With conditional independence assumption

# Naive Bayes Classifier

## ➤ Bayes rule:

$$\bullet P(Y = y_k | X_1, \dots, X_n) = \frac{P(Y=y_k)P(X_1, \dots, X_n | Y=y_k)}{\sum_j P(Y=y_j)P(X_1, \dots, X_n | Y = y_j)}$$

## ➤ Assume conditional independence among $X_i$ 's:

$$\bullet P(Y = y_k | X_1, \dots, X_n) = \frac{P(Y=y_k) \prod_i P(X_i | Y=y_k)}{\sum_j P(Y=y_j) \prod_i P(X_i | Y=y_j)}$$

## ➤ Pick the most probable (MAP) $Y$

$$\bullet \hat{Y} \leftarrow \operatorname{argmax}_{y_k} P(Y = y_k) \prod_i P(X_i | Y = y_k)$$

• New point is classified to  $Y_k$  if  $P(Y_k) \prod P(X_i | Y_k)$  is maximal.

## Example 1:

Suppose we got the new message with the words '**Dear Friend**', Decide whether this new message is a normal or spam message?

i.e. Normal/ Dear, Friend = Yes or No

Note if we know  $P(\text{Normal/ Dear, Friend})$  and  $P(\text{Spam/ Dear, Friend})$  then we can answer the question asked

| Email word | Spam | Email word | Spam |
|------------|------|------------|------|
| Dear       | Yes  | Friend     | No   |
| Friend     | No   | Friend     | Yes  |
| Dear       | No   | Dear       | No   |
| Dear       | No   | Lunch      | No   |
| Dear       | No   | Friend     | No   |
| Friend     | No   | Dear       | No   |
| Lunch      | No   | Dear       | No   |
| Friend     | No   | Dear       | No   |
| Lunch      | No   | Dear       | No   |
| Dear       | Yes  | Money      | Yes  |
| Money      | Yes  | Money      | No   |
| Money      | Yes  | Money      | Yes  |

## Example 1: Solution

**Step 1- View or collect “raw” data.**

**Step 2 - Convert long data to a frequency table**

|              | normal | spam | Row Total |
|--------------|--------|------|-----------|
| word         |        |      |           |
| Dear         | 8      | 2    | 10        |
| Friend       | 5      | 1    | 6         |
| Lunch        | 3      | 0    | 3         |
| Money        | 1      | 4    | 5         |
| Column Total | 17     | 7    | 24        |

**Step 3 - Row and column sums to get probabilities**

| Email word | Spam | Email word | Spam |
|------------|------|------------|------|
| Dear       | Yes  | Friend     | No   |
| Friend     | No   | Friend     | Yes  |
| Dear       | No   | Dear       | No   |
| Dear       | No   | Lunch      | No   |
| Dear       | No   | Friend     | No   |
| Friend     | No   | Dear       | No   |
| Lunch      | No   | Dear       | No   |
| Friend     | No   | Dear       | No   |
| Lunch      | No   | Dear       | No   |
| Dear       | Yes  | Money      | Yes  |
| Money      | Yes  | Money      | No   |
| Money      | Yes  | Money      | Yes  |

## Example 1: Solution

| Play         | normal | spam | Row Total |
|--------------|--------|------|-----------|
| word         |        |      |           |
| Dear         | 8      | 2    | 10        |
| Friend       | 5      | 1    | 6         |
| Lunch        | 3      | 0    | 3         |
| Money        | 1      | 4    | 5         |
| Column Total | 17     | 7    | 24        |

As  $P(N/D, F) > P(S/D, F)$ , we can decide that Dear Friend is Normal message.

**Step 4 - Apply probabilities from frequency table to Navie Bayes theorem**

$$P(N/D, F) = \frac{P(D, F/N) \cdot P(N)}{P(D, F)} = \frac{P(D/N) \cdot P(F/N) \cdot P(N)}{P(D/N) \cdot P(F/N) \cdot P(N) + P(D/S) \cdot P(F/S) \cdot P(S)} = \frac{\left(\frac{8}{17}\right) \cdot \left(\frac{5}{17}\right) \cdot \left(\frac{17}{24}\right)}{\left(\frac{8}{17}\right) \cdot \left(\frac{5}{17}\right) \cdot \left(\frac{17}{24}\right) + \left(\frac{2}{7}\right) \cdot \left(\frac{1}{7}\right) \cdot \left(\frac{7}{24}\right)} = \frac{0.098}{0.11} = 0.891$$

$$P(S/D, F) = \frac{P(D, F/S) \cdot P(S)}{P(D, F)} = \frac{P(D/S) \cdot P(F/S) \cdot P(S)}{P(D/N) \cdot P(F/N) \cdot P(N) + P(D/S) \cdot P(F/S) \cdot P(S)} = \frac{\left(\frac{2}{7}\right) \cdot \left(\frac{1}{7}\right) \cdot \left(\frac{7}{24}\right)}{\left(\frac{8}{17}\right) \cdot \left(\frac{5}{17}\right) \cdot \left(\frac{17}{24}\right) + \left(\frac{2}{7}\right) \cdot \left(\frac{1}{7}\right) \cdot \left(\frac{7}{24}\right)} = \frac{0.012}{0.11} = 0.109$$

## Example 1: Continued

Suppose we got the new message contains the word '**Lunch Money Money Money Money**', Decide whether this new message is a normal or spam message?

| Email word | Spam | Email word | Spam |
|------------|------|------------|------|
| Dear       | Yes  | Friend     | No   |
| Friend     | No   | Friend     | Yes  |
| Dear       | No   | Dear       | No   |
| Dear       | No   | Lunch      | No   |
| Dear       | No   | Friend     | No   |
| Friend     | No   | Dear       | No   |
| Lunch      | No   | Dear       | No   |
| Friend     | No   | Dear       | No   |
| Lunch      | No   | Dear       | No   |
| Dear       | Yes  | Money      | Yes  |
| Money      | Yes  | Money      | No   |
| Money      | Yes  | Money      | Yes  |

## Example 1: Solution

| Play         | normal | spam | Row Total |
|--------------|--------|------|-----------|
| word         |        |      |           |
| Dear         | 8      | 2    | 10        |
| Friend       | 5      | 1    | 6         |
| Lunch        | 3      | 0    | 3         |
| Money        | 1      | 4    | 5         |
| Column Total | 17     | 7    | 24        |

We can observe that we have to classify any message with Lunch as Normal message, no matter how many times we see the word Money and that's the problem.

**Issue with Naïve Bayes classifier:**  
**will not be able to classify as Normal or Spam!**

**Step 4 - Apply probabilities from frequency table**

$$P(N). P(L/N). P(M/N)^4 = \left(\frac{17}{24}\right) \cdot \left(\frac{3}{17}\right) \cdot \left(\frac{1}{17}\right)^4 = 0.0000015$$

$$P(S). P(L/S). P(M/S)^4 = \left(\frac{7}{24}\right) \cdot \left(\frac{0}{7}\right) \cdot \left(\frac{4}{7}\right)^4 = 0$$

To work around this problem add 1 count to the frequency table to each word. **(Laplace smoothing)**

## Example 1: Solution

| Play         | normal | spam | Row Total |
|--------------|--------|------|-----------|
| word         |        |      |           |
| Dear         | 8+1    | 2+1  | 12        |
| Friend       | 5+1    | 1+1  | 8         |
| Lunch        | 3+1    | 0+1  | 5         |
| Money        | 1+1    | 4+1  | 7         |
| Column Total | 21     | 11   | 32        |

As  $P(S/L, M^4) > P(N/L, M^4)$ , we can decide that **Lunch Money Money Money Money is Spam** message.

**Step 4 - Apply probabilities from frequency table**

$$P(N) \cdot P(L/N) \cdot P(M/N)^4 = \left(\frac{21}{32}\right) \cdot \left(\frac{4}{21}\right) \cdot \left(\frac{2}{21}\right)^4 = 0.00001$$

$$P(S) \cdot P(L/S) \cdot P(M/S)^4 = \left(\frac{11}{32}\right) \cdot \left(\frac{1}{11}\right) \cdot \left(\frac{5}{11}\right)^4 = 0.00133$$

## Example :2

If the feature of  
Today = (Outlook is Sunny ,  
Temp is Hot,  
Humidity is Normal,  
Windy is False),  
then the player will  
play or not?

| S. No | Outlook  | Temp | Humidity | Windy | Play Tennis |
|-------|----------|------|----------|-------|-------------|
| 1     | Rainy    | Hot  | High     | False | No          |
| 2     | Rainy    | Hot  | High     | True  | No          |
| 3     | Overcast | Hot  | High     | False | Yes         |
| 4     | Sunny    | Mild | High     | False | Yes         |
| 5     | Sunny    | Cool | Normal   | False | Yes         |
| 6     | Sunny    | Cool | Normal   | True  | No          |
| 7     | Overcast | Cool | Normal   | True  | Yes         |
| 8     | Rainy    | Mild | High     | False | No          |
| 9     | Rainy    | Cool | Normal   | False | Yes         |
| 10    | Sunny    | Mild | Normal   | False | Yes         |
| 11    | Rainy    | Mild | Normal   | True  | Yes         |
| 12    | Overcast | Mild | High     | True  | Yes         |
| 13    | Overcast | Hot  | Normal   | False | Yes         |
| 14    | Sunny    | Mild | High     | True  | No          |

## Example 2 Solution:

| Outlook  | Yes | No  |      |
|----------|-----|-----|------|
| Rainy    | 2/9 | 3/5 | 5/14 |
| Overcast | 4/9 | 0/5 | 4/14 |
| Sunny    | 3/9 | 2/5 | 5/14 |

| Humidity | Yes | No  |      |
|----------|-----|-----|------|
| High     | 3/9 | 4/5 | 7/14 |
| Normal   | 6/9 | 1/5 | 7/14 |

| Temp | Yes | No  |      |
|------|-----|-----|------|
| Hot  | 2/9 | 2/5 | 4/14 |
| Mild | 4/9 | 2/5 | 6/14 |
| Cool | 3/9 | 1/5 | 4/14 |

| Windy | Yes | No  |      |
|-------|-----|-----|------|
| True  | 3/9 | 3/5 | 6/14 |
| False | 6/9 | 2/5 | 8/14 |



## Example 2 Solution:

- $P(Yes) = \frac{9}{14}, P(No) = \frac{5}{14}$

Today = (Sunny, Hot, Normal, False)

Let Today = X, Sunny =  $X_1$ , Hot =  $X_2$ , Normal =  $X_3$ , False =  $X_4$

$$P(Yes/X) = \frac{P(X/Yes) P(Yes)}{P(X)} = \frac{P(X_1, X_2, X_3, X_4/Yes) P(Yes)}{P(X_1, X_2, X_3, X_4)} = \frac{P\left(\frac{X_1}{yes}\right) P\left(\frac{X_2}{yes}\right) P\left(\frac{X_3}{yes}\right) P\left(\frac{X_4}{yes}\right) \cdot P(Yes)}{P(X_1, X_2, X_3, X_4)}$$

Consider  $P\left(\frac{X_1}{yes}\right) P\left(\frac{X_2}{yes}\right) P\left(\frac{X_3}{yes}\right) P\left(\frac{X_4}{yes}\right) \cdot P(Yes) = \frac{3}{9} \times \frac{2}{9} \times \frac{6}{9} \times \frac{6}{9} \times \frac{9}{14} = 0.021$

Similarly for  $P(No/X)$

Consider  $P\left(\frac{X_1}{no}\right) P\left(\frac{X_2}{no}\right) P\left(\frac{X_3}{no}\right) P\left(\frac{X_4}{no}\right) \cdot P(no) = \frac{2}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{2}{5} \times \frac{5}{14} = 0.005$

$P(Yes/X) > P(No/X)$  . Hence, Player will play Tennis.

## Example:3

(a) Estimate Conditional Probability of each attributes {colour, legs, height, smelly} for the species classes: { S1, S2 } using the data given in the table.

(b) Using these probabilities in the table estimate the probabilities values for the new instance – { Colour =Green , Legs = 2, Height = Tall and Smelly= No}

| No | Colour | Legs | Height | Smelly | Species |
|----|--------|------|--------|--------|---------|
| 1  | White  | 3    | Short  | Yes    | S1      |
| 2  | Green  | 2    | Tall   | No     | S1      |
| 3  | Green  | 3    | Short  | Yes    | S1      |
| 4  | White  | 3    | Short  | Yes    | S1      |
| 5  | Green  | 2    | Short  | No     | S2      |
| 6  | White  | 2    | Tall   | No     | S2      |
| 7  | White  | 2    | Tall   | No     | S2      |
| 8  | White  | 2    | Short  | Yes    | S2      |

## Example 3 Solution:

$$P(S1) = \frac{4}{8}, P(S2) = \frac{4}{8},$$

| Colour | S1  | S2  |
|--------|-----|-----|
| White  | 2/4 | 3/4 |
| Green  | 2/4 | 1/4 |

| Height | S1  | S2  |
|--------|-----|-----|
| Tall   | 3/4 | 2/4 |
| Short  | 1/4 | 2/4 |



| Legs | S1  | S2  |
|------|-----|-----|
| 2    | 1/4 | 4/4 |
| 3    | 3/4 | 0/4 |

| Smelly | S1  | S2  |
|--------|-----|-----|
| Yes    | 3/4 | 1/4 |
| No     | 1/4 | 3/4 |

## Example 3 Solution:

$$\begin{aligned}P\left(\frac{S1}{\text{New Instance}}\right) &= P(S1) \times P\left(\frac{\text{Colour=Green}}{S1}\right) \times P\left(\frac{\text{Leg=2}}{S1}\right) \times P\left(\frac{\text{Height=Tall}}{S1}\right) \times \frac{P(\text{Smelly=no})}{S1} \\&= 1/2 \times 2/4 \times 1/4 \times 3/4 \times 1/4 = 0.0117\end{aligned}$$

$$\begin{aligned}P\left(\frac{S2}{\text{New Instance}}\right) &= P(S2) \times P\left(\frac{\text{Colour=Green}}{S2}\right) \times P\left(\frac{\text{Leg=2}}{S2}\right) \times P\left(\frac{\text{Height=Tall}}{S2}\right) \times \frac{P(\text{Smelly=no})}{S2} \\&= 1/2 \times 1/4 \times 4/4 \times 2/4 \times 3/4 = 0.047\end{aligned}$$

$$P\left(\frac{S2}{\text{New Instance}}\right) > P\left(\frac{S1}{\text{New Instance}}\right).$$

Hence, New instance belongs to species S2

# Naïve Bayes for Text Classification

- Naïve Bayes is commonly used for **text classification**
- For a document with  $k$  terms  $d = (t_1, \dots, t_k)$

Fraction of documents in  $c$

$$P(c|d) = P(c)P(d|c) = P(c) \prod_{t_i \in d} P(t_i|c)$$

- $P(t_i|c)$  = Fraction of terms from **all documents** in  $c$  that are  $t_i$ .

Number of times  $t_i$  appears in some document in  $c$

$$P(t_i|c) = \frac{N_{ic} + 1}{N_c + T}$$

Laplace Smoothing

Total number of terms in all documents in  $c$

Number of unique words (vocabulary size)

- Easy to implement and works relatively well
- **Limitation**: Hard to incorporate **additional features** (beyond words).
  - E.g., number of adjectives used.

## Example 4:

| Text                           | Tag        |
|--------------------------------|------------|
| "A great game"                 | Sports     |
| "The election was over"        | Not sports |
| "Very clean match"             | Sports     |
| "A clean but forgettable game" | Sports     |
| "It was a close election"      | Not sports |

Which tag does the sentence *A very close game* belong to? i.e.  $P(\text{sports} | \text{A very close game})$

Feature Engineering: Bag of words i.e use word frequencies without considering order

Using Bayes Theorem:

$P(\text{sports} | \text{A very close game})$

$= \frac{P(\text{A very close game} | \text{sports}) P(\text{sports})}{P(\text{A very close game})}$

We assume that every word in a sentence is **independent** of the other ones

"close" doesn't appear in sentences of sports tag, So  $P(\text{close} | \text{sports}) = 0$ , which makes product 0

## Example 4 Solution: Apply Laplace smoothing

| Text                           | Tag        |
|--------------------------------|------------|
| "A great game"                 | Sports     |
| "The election was over"        | Not sports |
| "Very clean match"             | Sports     |
| "A clean but forgettable game" | Sports     |
| "It was a close election"      | Not sports |

| Word  | P(Word/Sport) | P(Word/Not Sport) |
|-------|---------------|-------------------|
| a     | 2/11          | 1/9               |
| very  | 1/11          | 0/9               |
| close | 0/11          | 1/9               |
| game  | 2/11          | 0/9               |

## Example 4 Solution:

| Text                           | Tag        |
|--------------------------------|------------|
| "A great game"                 | Sports     |
| "The election was over"        | Not sports |
| "Very clean match"             | Sports     |
| "A clean but forgettable game" | Sports     |
| "It was a close election"      | Not sports |

Which tag does the sentence *A very close game* belong to? i.e.  $P(\text{sports} \mid \text{A very close game})$

Using Bayes Theorem:

$$\begin{aligned} P(\text{sports} \mid \text{A very close game}) \\ &= \frac{P(\text{A very close game} \mid \text{sports}) P(\text{sports})}{P(\text{A very close game})} \end{aligned}$$

We assume that every word in a sentence is **independent** of the other ones

$$\begin{aligned} P(\text{A very close game}) &= P(\text{A}) P(\text{very}) P(\text{close}) P(\text{game}) \\ P(\text{A very close game} \mid \text{sports}) &= P(\text{a} \mid \text{sports}) P(\text{very} \mid \text{sports}) \\ &\quad P(\text{close} \mid \text{sports}) P(\text{game} \mid \text{sports}) \end{aligned}$$

"close" doesn't appear in sentences of sports tag, So  $P(\text{close} \mid \text{sports}) = 0$ , which makes product 0

## Example 4 Solution: Laplace smoothing

- Laplace smoothing: we add 1 or in general constant  $k$  to every count so it's never zero.
- To balance this, we add the number of possible words to the divisor, so the division will never be greater than 1
- In our case, the possible words are ['a', 'great', 'very', 'over', 'it', 'but', 'game', 'election', 'clean', 'close', 'the', 'was', 'forgettable', 'match'].

## Example 4 Solution: Apply Laplace smoothing

| Word  | P(word   Sports) | P(word   Not Sports) |
|-------|------------------|----------------------|
| a     | $2+1 / 11+14$    | $1+1 / 9+14$         |
| very  | $1+1 / 11+14$    | $0+1 / 9+14$         |
| close | $0+1 / 11+14$    | $1+1 / 9+14$         |
| game  | $2+1 / 11+14$    | $0+1 / 9+14$         |

$$\begin{aligned} &P(a|Sports) \times P(very|Sports) \times P(close|Sports) \times P(game|Sports) \times \\ &P(Sports) \\ &= 2.76 \times 10^{-5} \\ &= 0.0000276 \end{aligned}$$

$$\begin{aligned} &P(a|Not Sports) \times P(very|Not Sports) \times P(close|Not Sports) \times \\ &P(game|Not Sports) \times P(Not Sports) \\ &= 0.572 \times 10^{-5} \\ &= 0.00000572 \end{aligned}$$

## Example 5:

| Doc No | Text                        |            |
|--------|-----------------------------|------------|
| 1      | I LOVED THE MOVIE           | _ POSITIVE |
| 2      | I HATED THE MOVIE           | _ NEGATIVE |
| 3      | A GREAT MOVIE ,GOOD MOVIE   | _ POSITIVE |
| 4      | POOR ACTING                 | _ NEGATIVE |
| 5      | GREAT ACTING , A GOOD MOVIE | _ POSITIVE |
| NEW    | I HATED THE POOR ACTING     | _????      |

## Example 5 Solution:

$$P(c/x)$$

$$P(+ / \text{I hated the poor acting}) =$$

$$P(- / \text{I hated the poor acting}) =$$

Based on these probabilities, we can decide the class which the new text belongs

## Example 5 Solution:

$P(+ \mid \text{I hated the acting})$

$$\text{ie } P(c_i \mid x) = \frac{P(x \mid c_i) P(c_i)}{P(x)}$$

$$= P(\text{I} \mid +) P(\text{hated} \mid +) P(\text{the} \mid +) P(\text{acting} \mid +) P(+)$$

$$\frac{P(\text{I}, +)}{P(+)}$$

$$\frac{P(\text{hated}, +)}{P(+)}$$

## Example 5 Solution:

| words  | positive | negative |
|--------|----------|----------|
| I      | 1        | 1        |
| loved  | 1        | 0        |
| the    | 1        | 1        |
| movie  | 4        | 1        |
| hated  | 0        | 1        |
| a      | 2        | 0        |
| great  | 2        | 0        |
| poor   | 0        | 1        |
| acting | 1        | 1        |
| good   | 2        | 0        |

$$P(\pm / +)$$

$$= \frac{1 + 1}{14 + 10}$$

$$P(I / -)$$

$$= \frac{1 + 1}{6 + 10}$$

## Example 5 Solution:

"I hated the poor acting"

word

positive

negative

I

$$\frac{1+1}{14+10} = 0.0833$$

$$\frac{1+1}{6+10} = 0.125$$

hated

$$\frac{0+1}{14+10} = 0.0417$$

$$\frac{1+1}{6+10} = 0.125$$

the

$$\frac{1+1}{14+10} = 0.0833$$

$$\frac{1+1}{6+10} = 0.125$$

poor

$$\frac{0+1}{14+10} = 0.0417$$

$$\frac{1+1}{6+10} = 0.125$$

acting

$$\frac{1+1}{14+10} = 0.0833$$

$$\frac{1+1}{6+10} = 0.125$$

## Example 5 Solution:

"I hated the poor acting"

word

positive

negative

I

$$\frac{1+1}{14+10} = 0.0833$$

$$\frac{1+1}{6+10} = 0.125$$

hated

$$\frac{0+1}{14+10} = 0.0417$$

$$\frac{1+1}{6+10} = 0.125$$

the

$$\frac{1+1}{14+10} = 0.0833$$

$$\frac{1+1}{6+10} = 0.125$$

poor

$$\frac{0+1}{14+10} = 0.0417$$

$$\frac{1+1}{6+10} = 0.125$$

acting

$$\frac{1+1}{14+10} = 0.0833$$

$$\frac{1+1}{6+10} = 0.125$$

$$P(+ / x)$$

$$= (0.083) (0.0417) (0.083) (0.0417) (0.083) \times P(+)$$

$\downarrow$   
 $3/5$

$$= 6.03 \times 10^{-7} \quad \checkmark$$

$$P(- / x)$$

$$= ( ) ( ) ( ) ( ) ( ) ( ) P(-)$$

$\downarrow$   
 $2/8$

$$= 1.22 \times 10^{-5}$$

$\therefore$  Negative class

# Naïve Bayes Classifier Applications

## Categorizing News



## BUSINESS & ECONOMY

### Paying service charge at hotels not mandatory



TECHNOLOGY &amp; SCIENCE

## The 'dangers' of being admin of a WhatsApp group



ENTERTAINMENT

This actor stars in Raabta.  
Guess who?



IPL 2017

### Preview: Bullish KKR face depleted Lions



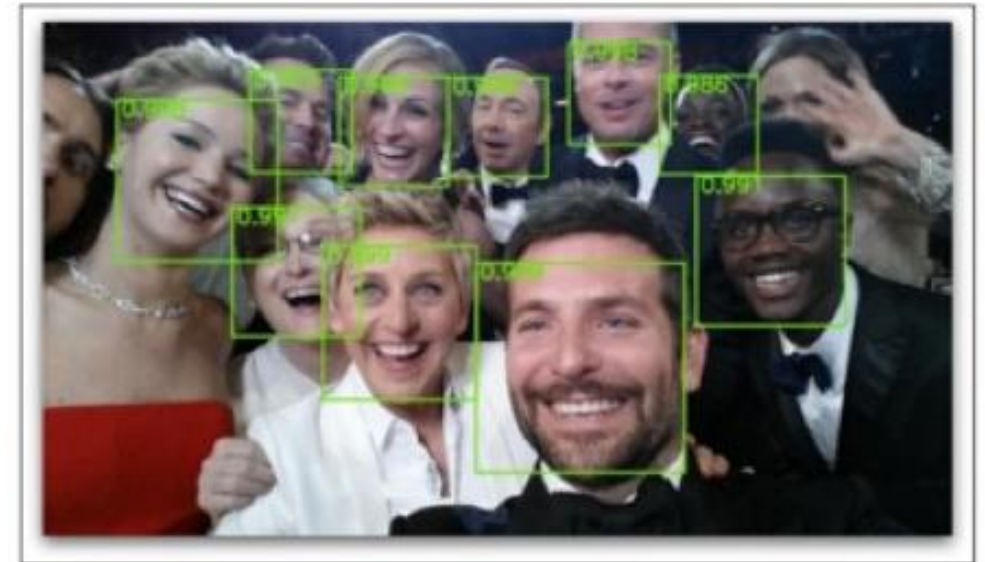
## INDIA

### Why is Aadhaar mandatory for PAN? SC asks Centre

## Email Spam Detection



## Face Recognition



## Sentiment Analysis



# Practical Issues of Bayesian learning

- ✓ Along with decision trees, neural networks, one of the most practical learning methods.
- ✓ When to use
  - ✓ Moderate or large training set available
  - ✓ Attributes that describe instances are conditionally independent given classification
- ✓ Successful applications:
  - ✓ Diagnosis
  - ✓ Classifying text documents

## Practical Issues of Bayesian learning

- Require initial knowledge of many probabilities
  - Often estimated based on background knowledge, previously available data, and assumptions about the form of the underlying distributions.
- Significant computational cost required to determine the Bayes optimal hypothesis in the general case (linear in the number of candidate hypotheses)

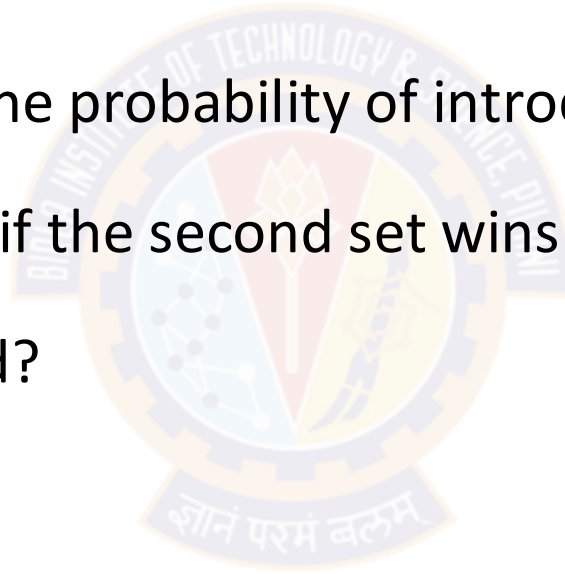
## Glossary

- Bayes theorem
- MAP Hypothesis
- ML Estimate
- Naïve Bayes theorem
- NB with Laplace correction



# Practice Problems 1

Two set of candidates are competing for the positions on the board of directors of a company. The probabilities that the first and the second sets will win are 0.6 and 0.4 respectively. If the first set wins , the probability of introducing a new product is 0.8, and the corresponding probability if the second set wins is 0.3. What is the probability that the product will be introduced?



# Solution

Let  $E_1$  ,  $E_2$  denote the events that the First and the second sets of candidates win and  $E$  be the Event of introducing a new product.

$$P(E_1) = 0.6; P(E_2) = 0.4; P(E/E_1) = 0.8; P(E/E_2) = 0.3;$$

By Addition theorem of probability

$$\begin{aligned} P(E) &= P(E_1 \cap E) + P(E_2 \cap E) = P(E_1)P\left(\frac{E}{E_1}\right) + P(E_2)P\left(\frac{E}{E_2}\right) = 0.6 \times 0.8 + 0.4 \times 0.3 \\ &= 0.48 + 0.12 = 0.6. \end{aligned}$$

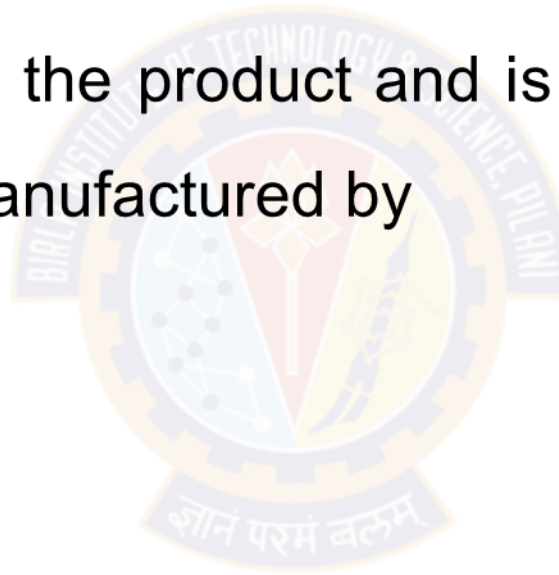
## Practice Problems 2

In a bolt factory, Machines A, B, C manufacture respectively 25%, 35% and 40 % of the total. Of their output 5, 4, 2 percent are known to be defective bolts.

A bolt is drawn at random from the product and is found to be defective. What are the probabilities that it was manufactured by

(i) Machine A

(ii) Machine B or C



# Solution

Let  $E_1, E_2$  and  $E_3$  denote respectively the events that the bolt selected at random is manufactured by the machines A, B and C respectively.

Let E denote the event that it is defective.

(i) The probability that the defective bolt is manufactured by Machine A is:

$$P(E_1) = 0.25, P(E_2) = 0.35, P(E_3) = 0.40$$

$$P(E/E_1) = 0.05, P(E/E_2) = 0.04, P(E/E_3) = 0.02,$$

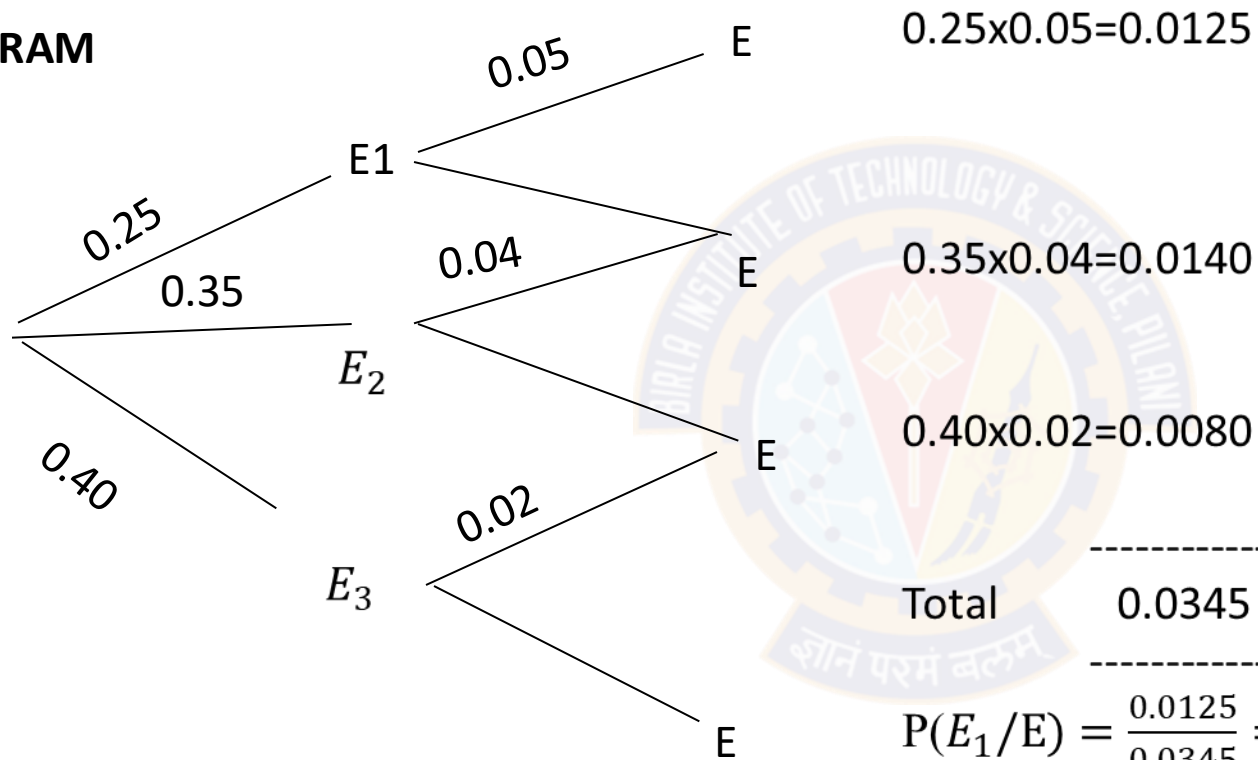
$$P(E) = 0.25 \times 0.05 + 0.35 \times 0.04 + 0.40 \times 0.02 = 0.0345.$$

$$P(E_1/E) = \frac{P(E_1)P(\frac{E}{E_1})}{\sum P(E_i)P(\frac{E}{E_i})} = \frac{0.0125}{0.0345} = 0.36$$

(ii) Similarly,  $P(E_2/E) = 0.41$ ,  $P(E_3/E) = 0.23$ , Hence Defective bolt chosen at random is Manufactured by B or C =  $P(E_2/E) + P(E_3/E) = 0.41 + 0.23 = 0.64$ .

# Conti..

## TREE DIAGRAM



$$P(E_1/E) = \frac{0.0125}{0.0345} = 0.36, P(E_2/E) = 0.41, \\ P(E_3/E) = 0.23$$

# Practice Problems 3

The results of an investigation by an expert on a fire accident in a skyscraper are summarized below:

- (i)  $P(\text{There should have been short circuit})=0.8$
- (ii)  $P(\text{LPG cylinder explosion})$
- (iii) Chance of fire accident is 30% given a short circuit and 95% given an LPG explosion.

Based on these, What do you think is the most probable causes of fire? Statistically justify your answer.

# Solution

Let  $E_1$  : Short circuit ;  $E_2$ : *LPG explosion* ;  $E$ : Fire accident

Then,

$$P(E_1) = 0.8 ; P(E_2) = 0.2 ; P(E/E_1) = 0.30 ; P(E/E_2) = 0.95$$

By Baye's Rule:

$$P(E_1/E) = \frac{P(E_1) \cdot P(E/E_1)}{P(E_1) \cdot P(E/E_1) + P(E_2) \cdot P(E/E_2)} = \frac{0.8 \times 0.30}{0.8 \times 0.30 + 0.2 \times 0.95} = \frac{0.240}{0.240 + 0.190} = \frac{24}{43}$$

$$P(E_2/E) = \frac{P(E_2) \cdot P(E/E_2)}{P(E_1) \cdot P(E/E_1) + P(E_2) \cdot P(E/E_2)} = \frac{0.190}{0.430} = \frac{19}{43}$$

(OR)

$$P(E_2/E) = 1 - P(E_1/E) = 1 - \frac{24}{43} = \frac{19}{43}$$

# Practice Problems 4

- The chances that an academician, a business man and a politician becoming Vice Chancellor of an university are 0.5, 0.3 and 0.2 respectively. The probability that research work will be promoted in the university by these 3 gentlemen are respectively are 0.8, 0.6 and 0.4. It is found Research work has been promoted by the university. What is the chance that an academician has become the VC?

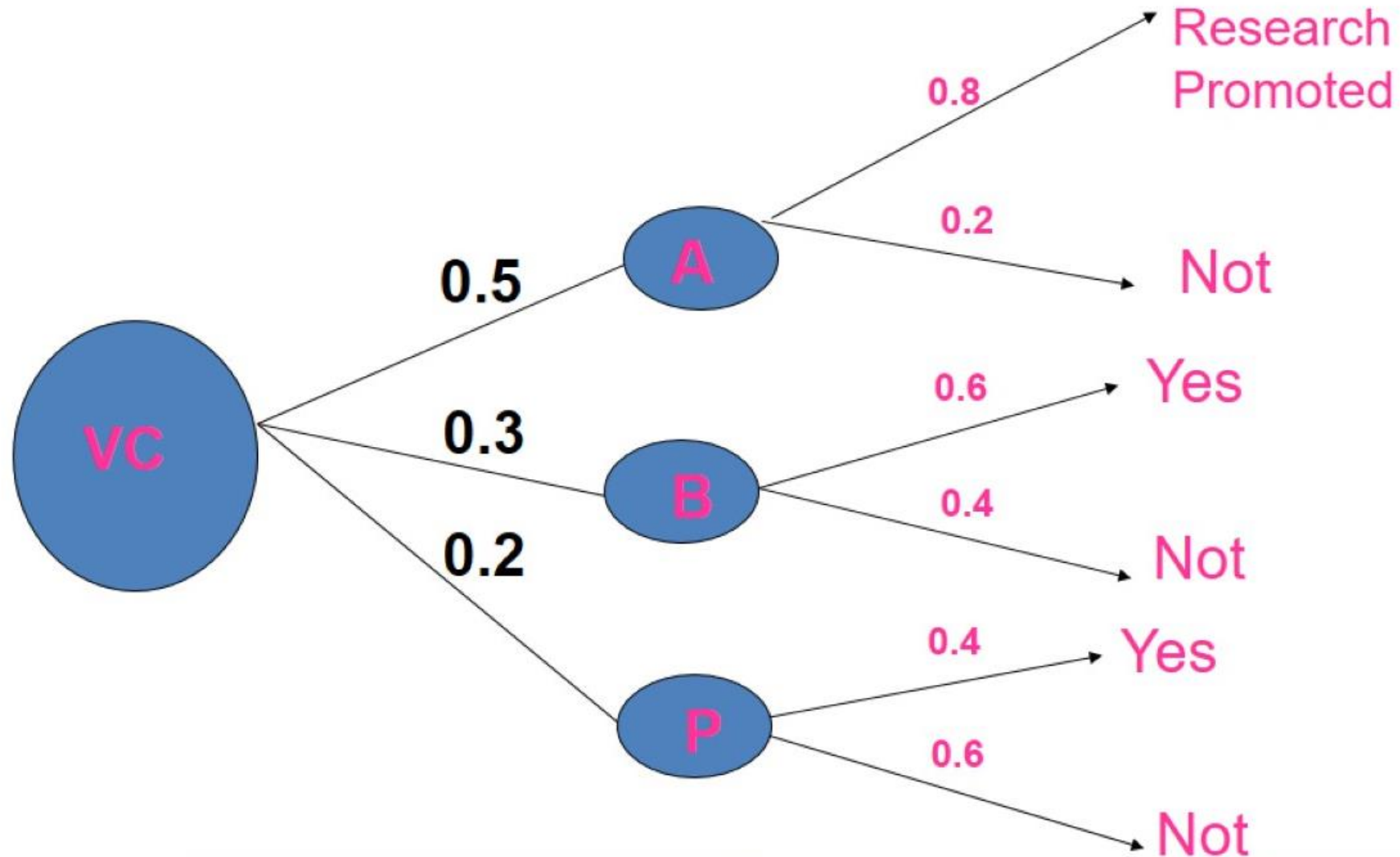
- Solution:**

Let **X: “ Research work is promoted”**

Clearly,  $P(X) = 0.5 \times 0.8 + 0.3 \times 0.6 + 0.2 \times 0.4 = 0.66$

Now to find  $P[\text{“ An Academician is VC”} / \text{“Research work is promoted i.e. event X”}] = \frac{0.5 \times 0.8}{0.66} = 0.6061$

## Conti..



# Practice Problems 5

Four technicians regularly make repairs when breakdowns occur on an automated production line.

- Janet, who services 20% of the breakdowns, makes an incomplete repair 1 time in 20;
- Tom, who services 60% of the breakdowns, makes an incomplete repair 1 time in 10;
- Georgia, who services 15% of the breakdowns, makes an incomplete repair 1 time in 10;
- Peter, who services 5% of the breakdowns, makes an incomplete repair 1 time in 20.

For the next problem with the production line diagnosed as being due to an initial repair that was incomplete, what is the probability that this initial repair was made by Janet?

# Solution

Let  $A$  be the event that the initial repair was incomplete,  $B_1$  that the initial repair was made by Janet,  $B_2$  that it was made by Tom,  $B_3$  that it was made by Georgia. and  $B_4$  that it was made by Peter.

$$P(B_1|A) = \frac{(0.20)(0.05)}{(0.20)(0.05) + (0.60)(0.10) + (0.15)(0.10) + (0.05)(0.05)}$$
$$= \mathbf{0.114}$$

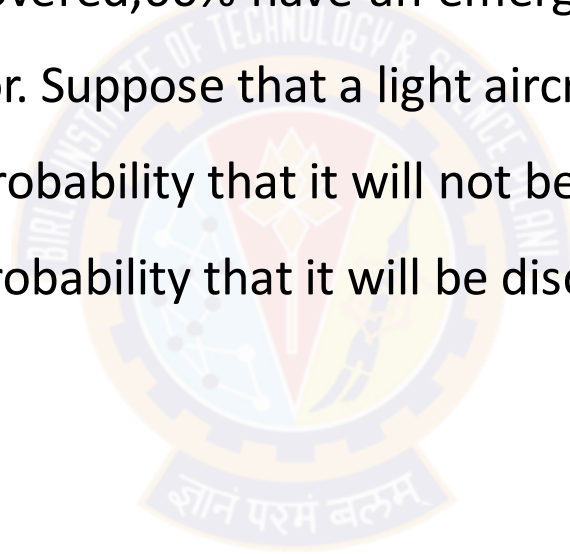
and it is of interest to note that although Janet makes an incomplete repair only 1 out of 20 times, namely, 5% of the breakdowns she services, more than 11% of the incomplete repairs are her responsibility.

## Practice Problems 6

Seventy percent of the light aircraft that disappear while in flight in a certain country are subsequently discovered. Of the aircraft that are discovered, 60% have an emergency locator, whereas 90% of the aircraft not discovered do not have such a locator. Suppose that a light aircraft has disappeared.

If it has an emergency locator, find the probability that it will not be discovered.

(b) If it doesn't have a locator, find the probability that it will be discovered.



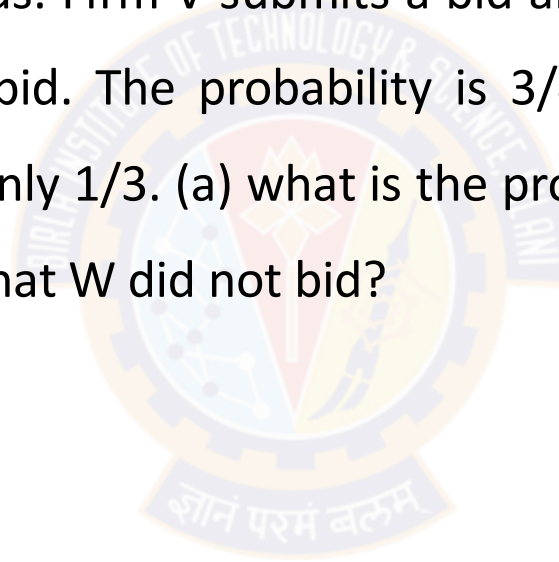
# Solution

The tree diagram below shows the probability for the four disjoint options;

$$\begin{aligned} &P(\text{the flight is discovered and has a locator}) \\ &= P(\text{discovered})P(\text{locator} \mid \text{discovered}) \\ &= (.7)(.6) = .42. \end{aligned}$$

## Practice Problems 7

Two firms V and W consider bidding on a road-building job, which may or may not be awarded depending on the amounts of the bids. Firm V submits a bid and the probability is  $\frac{3}{4}$  that it will get the job provided firm W does not bid. The probability is  $\frac{3}{4}$  that W will bid, and if it does, the probability that V will get the job is only  $\frac{1}{3}$ . (a) what is the probability that V will get the job? (b) If V gets the job, what is the probability that W did not bid?



# Solution

Given:  $P(V/W')=3/4$ ,  $P(W) = 1/3$ ,  $P(V/W)=1/3$ ,  $P(W') = 2/3$

$$(a) V = (V \cap W) \cup (V \cap W') = P(V) = P(V \cap W) + P(V \cap W')$$

$$P(V) = P(V/W) P(W) + P(V/W') P(W') = 11/18$$

$$(b) P(W'/V) = P(V/W') P(W') / P(V) = 9/11$$



Thank you